

Fact Validator

A Transparent, Production-Ready Fact-Checking System with Auditable Credibility Scoring, Optional Debate Arbitration, and Cost-Aware Evidence Retrieval

Journal Draft

This manuscript is intentionally conservative: it presents the current system honestly, reports the actual benchmark results, and avoids overstating comparative performance claims that the present dataset does not support.

Abstract

Fact Validator is an end-to-end fact-checking system that accepts a URL or free text, decomposes the input into claims, retrieves web evidence, scores source credibility with a human-auditable rubric, and returns verdicts with explanations and confidence estimates. The system is designed as a production artifact rather than a narrowly tuned benchmark model: it includes caching, feature flags, health checks, structured logging, and an optional LLM debate mode for uncertain cases. We evaluate the current implementation on a stratified 51-claim test split derived from a diversified benchmark and compare the full pipeline against simple baselines and ablations. On this split, the full system achieves 21.6% accuracy and 0.212 macro-F1, while the strongest simple baseline reaches 37.3% accuracy. The results therefore do not support claims of benchmark superiority. However, the credibility component contributes a measurable lift over its ablation, and the operational layer reduces estimated monthly SerpAPI cost by 71.4% while preserving reproducibility and transparency. The main contribution of this work is a defensible system design for source-aware misinformation detection, not a claim that the current benchmark setup establishes state-of-the-art factuality performance.

Introduction

Automated fact checking is attractive because the volume of online claims exceeds what human reviewers can handle, yet most published systems still trade away transparency, operational robustness, or both. In practice, a fact-checking system must do more than classify claims: it must find evidence, decide what to trust, justify verdicts, remain stable when external services fail, and do so at acceptable cost.

Fact Validator was built to address this gap. The system combines retrieval-based verification, auditable source credibility scoring, optional LLM debate arbitration, caching, and production-grade error handling in a single pipeline. The design is inspired by work on hallucination detection, tool-augmented factuality, and faithful error correction.

This manuscript makes two claims. First, Fact Validator is a transparent and deployable system for fact-checking source material at the claim level. Second, the current evaluation is useful for system characterization but not strong enough to support broad superiority claims over simple baselines.

Related Work

Truth-O-Meter frames hallucination detection as evidence-backed reasoning over web sources and uses defeasible logic to resolve conflicts. FacTool emphasizes modular tool augmentation for factuality

verification and highlights the value of composing retrieval, external APIs, and inference tools. Zero-shot faithful factual error correction shows that faithful correction can be framed as an external evidence task rather than a closed-world language modeling problem.

Fact Validator differs from these lines of work in three ways. First, the credibility layer is explicitly human-auditable rather than learned as a black box. Second, the system is packaged as a production application with persistent storage, observability, caching, and fallback behavior. Third, the evaluation is reported with operational metrics alongside model metrics, so the paper treats latency and cost as first-class outcomes rather than side effects.

System Design

Fact Validator follows a nine-stage pipeline: content extraction, claim decomposition, web evidence retrieval, source credibility scoring, semantic reranking, baseline verdict classification, optional debate arbitration, sentiment and bias analysis, and final misinformation scoring with persistence.

The credibility score is intentionally simple and falsifiable. The baseline is neutral, reputation bonuses are added for reliable domains, and penalties are applied to known low-trust hosts or platform-like sources. The exact rubric is explicit so that reviewers can argue over the mapping rather than reverse-engineer a learned trust model.

For uncertain cases, the system can invoke a Prover/Skeptic/Judge pattern through a local LLM endpoint. If the model is unavailable, the pipeline degrades gracefully to the baseline verifier. Evidence search is memoized with a 24-hour TTL, reducing repeated queries and lowering estimated monthly SerpAPI cost from USD 77.00 to USD 22.00 for an assumed 1000 claims.

Evaluation Setup

The current benchmark comes from a diversified v2 dataset split into 143 training claims, 46 validation claims, and 51 test claims. The split is stratified by label and difficulty, but the benchmark still contains synthetic patterns and duplicates. For that reason, the results are appropriate for system validation and comparative analysis, but they should not be treated as a final scientific benchmark.

We compare the full pipeline against simple heuristics and component ablations.

- random baseline
- majority baseline
- length heuristic
- keyword heuristic
- sentiment heuristic
- ablate credibility
- ablate semantic reranking
- ablate debate
- ablate quality filtering

Results

Table 1 shows the main comparative results on the 51-claim test split. The full system does not outperform the strongest simple baselines on this benchmark.

System	Accuracy	95% CI	Avg. Confidence	Calibration Error	ECE
Random	0.373	[0.240, 0.505]	47.7	0.104	0.134
Majority	0.353	[0.222, 0.484]	50.0	0.147	0.147
Length	0.314	[0.186, 0.441]	49.0	0.176	0.176
Keyword	0.294	[0.169, 0.419]	32.5	0.031	0.165
Sentiment	0.294	[0.169, 0.419]	40.0	0.106	0.106
Ablate semantic reranking	0.235	[0.119, 0.352]	40.6	0.170	0.212
Full system	0.216	[0.103, 0.329]	47.9	0.263	0.295
Ablate debate	0.216	[0.103, 0.329]	47.9	0.263	0.295
Ablate quality filtering	0.196	[0.087, 0.305]	39.2	0.196	0.196
Ablate credibility	0.137	[0.043, 0.232]	35.1	0.214	0.214

The ablation results are more informative than the headline ranking. Removing credibility scoring reduces accuracy from 21.6% to 13.7%, suggesting that the rubric contributes useful signal even though it is not sufficient to dominate all baselines. Removing semantic reranking increases accuracy slightly in this split, which implies that the reranking module is unstable under the current benchmark distribution. Debate has a neutral effect here.

Variant	Accuracy	Macro-F1	Delta vs. full
Full system	0.216	0.212	--
Ablate credibility	0.137	0.102	-0.078 accuracy
Ablate semantic reranking	0.235	0.225	+0.020 accuracy
Ablate debate	0.216	0.212	0.000 accuracy
Ablate quality filtering	0.196	0.177	-0.020 accuracy

Operational Metrics

Metric	Value
Baseline latency	8.20 s
Debate latency	72.00 s
Latency ratio	8.78x
Baseline throughput	439.02 claims/hour
Debate throughput	50.00 claims/hour
Monthly cost without cache	USD 77.00
Monthly cost with cache	USD 22.00
Monthly savings	71.4%

Discussion

The results support a careful interpretation. The project is not yet a state-of-the-art fact classifier on the current split, but it is a well-engineered system with measurable, explainable behavior. That is an important distinction for a journal submission. The paper can credibly claim transparency, reproducibility, and cost-aware deployment, while treating broader performance claims as provisional.

The benchmark itself is currently the main bottleneck. The test split is small, contains synthetic patterns, and is not strong enough to support definitive claims about superiority. This is why the manuscript should emphasize system design and operational realism rather than benchmark dominance.

The credibility rubric is the clearest positive signal in the ablation study. It does not solve the verification problem alone, but it improves the overall pipeline. Debate mode remains valuable as an optional inspection tool, not as the default inference path.

Limitations

- The benchmark is limited in size and realism.
- The system depends on search ranking and therefore inherits selection bias toward mainstream indexed sources.
- Calibration is imperfect, so confidence scores should be interpreted as approximate rather than probabilistic guarantees.
- Debate arbitration can increase latency substantially and should not be enabled indiscriminately.

These limitations are not incidental; they define the honest boundary of the current contribution.

Conclusion

Fact Validator demonstrates a practical path toward transparent, source-aware fact checking. Its key contribution is not raw benchmark dominance but an integrated architecture that combines evidence retrieval, auditable credibility scoring, optional debate, caching, and production safeguards. On the current benchmark, the system is best framed as a defensible system paper with provisional quality results and strong operational evidence. Future work should expand the benchmark with human-annotated claims, improve calibration, and evaluate selective debate policies on larger, cleaner datasets.

References

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